

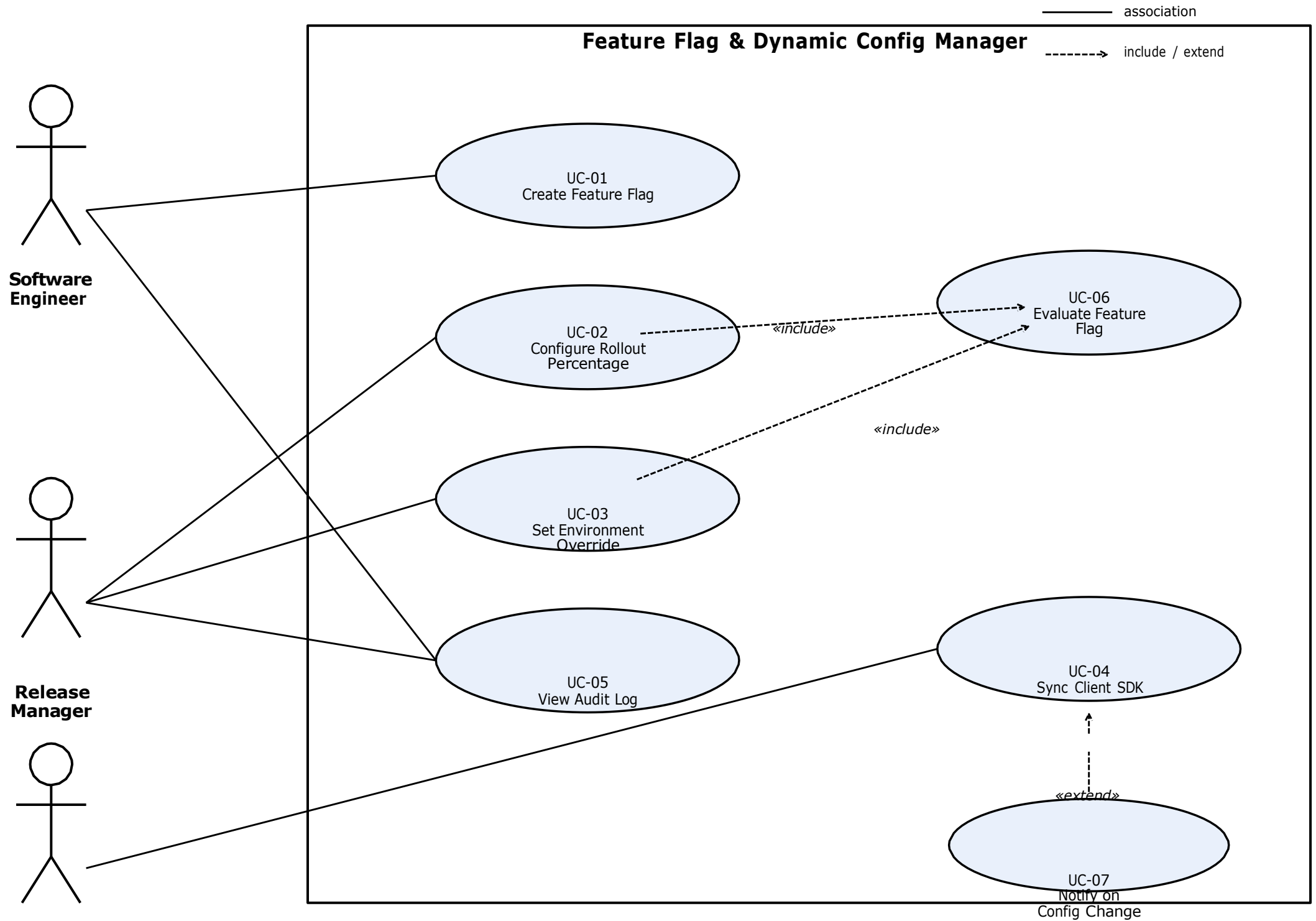
Lab 1 – Requirements Table

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Problem Statement #43: Feature Flag & Dynamic Config Manager

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall evaluate user targeting rules (e.g., User ID hash, Beta cohort) and return accurate boolean feature flag states.	High	Pass: Target 10% rollout cohort receives enabled flag. Fail: User outside cohort receives enabled flag.	Core functionality of a feature flag engine; incorrect targeting undermines the whole system.
FR-002	Functional	The system shall allow a Software Engineer to create a new feature flag with a unique key and a default (on/off) state.	High	Pass: Submitting a flag key that does not already exist creates the flag and it appears in the flag list within 2 seconds. Fail: Duplicate key is rejected with an error.	Engineers need a reliable way to register new flags before they can be targeted or rolled out.
FR-003	Functional	The system shall allow a Release Manager to configure a percentage-based rollout (0-100%) for an existing feature flag.	High	Pass: Setting rollout to 25% results in approximately 25% ($\pm 2\%$) of a 10,000-user sample receiving the enabled flag over 3 test runs. Fail: Actual exposure deviates by more than 2%.	Percentage rollouts let teams de-risk releases by exposing a new feature gradually.
FR-004	Functional	The system shall allow a Release Manager to set environment-specific overrides (Dev, Staging, Prod) that take precedence over the default rollout rule.	Medium	Pass: Setting an override of "ON" for Staging returns the flag as enabled for all Staging requests regardless of rollout percentage. Fail: Default rollout percentage is applied instead of the override.	Teams need to force a flag on/off in a specific environment for testing without affecting production users.
FR-005	Functional	The system shall synchronize any feature flag state change to all connected client SDKs within 5 seconds of the change being saved.	High	Pass: A flag toggle triggers an update event received by a connected test SDK client within 5 seconds. Fail: SDK client still reflects the stale value after 5 seconds.	Real-time sync ensures users see consistent behavior and enables fast kill-switch rollbacks.
NFR-001	Nonfunctional	The feature flag evaluation API must execute in under 15 ms to minimize overhead when integrated into client critical paths.	High	Pass: Benchmarking tests confirm P95 latency under 15 ms under simulated peak load (1000 req/s). Fail: P95 latency exceeds 15 ms.	Flag evaluation sits on the hot path of client applications; slow evaluation degrades the host application's performance.
NFR-002	Nonfunctional	The system shall ensure that only authenticated users with the Release Manager role can modify rollout percentages or environment overrides.	High	Pass: An authenticated Software Engineer (non-Release-Manager) attempting to change a rollout percentage receives a 403 Forbidden response. Fail: The change is accepted.	Uncontrolled access to rollout controls could expose unfinished features to all users or disable production functionality.



Client SDK

Lab 1 – Use-Case Flow Specification

Problem Statement #43: Feature Flag & Dynamic Config Manager

Use Case: UC-02 – Configure Rollout Percentage

Primary Actor: Release Manager

Supporting Actors: System (flag evaluation engine), Client SDKs

Preconditions

- The Release Manager is authenticated and has the Release Manager role (see NFR-002).
- The target feature flag already exists in the system (created via UC-01).

Postconditions

- The feature flag's rollout percentage is updated and persisted.
- All connected client SDKs reflect the new rollout percentage within 5 seconds (FR-005).
- An entry describing the change is written to the audit log.

Main Success Scenario

- 1 The Release Manager selects a feature flag from the flag dashboard.
- 2 The system displays the flag's current rollout configuration (percentage, targeting rules, environment overrides).
- 3 The Release Manager enters a new rollout percentage between 0 and 100.
- 4 The system validates that the entered value is a number within the 0–100 range.
- 5 The system evaluates the flag's existing targeting rules to confirm the new percentage is compatible «include» UC-06 – Evaluate Feature Flag.
- 6 The system saves the updated rollout configuration.
- 7 The system synchronizes the updated configuration to all connected client SDKs.
- 8 The system writes an entry to the audit log recording the old value, new value, timestamp, and Release Manager identity.
- 9 The system displays a confirmation message to the Release Manager. Use case ends successfully.

Alternate Flow – 4a. Invalid Percentage Entered

- a The system detects that the entered value is not a number, or is outside the 0–100 range.
- b The system displays a validation error message and does not save the change.
- c The Release Manager is prompted to re-enter a valid percentage.
- d If three consecutive invalid attempts are made, the system cancels the configuration action and logs the cancelled attempt to the audit log.